

Credit Rationing in Markets with Imperfect Information

Joseph E. Stigitz & Andrew Weiss (1981)

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Motivation

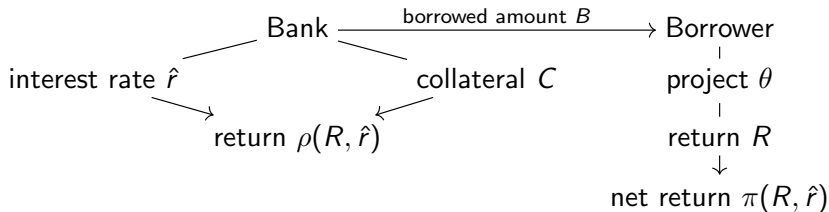
- ▶ **The Core Objective:**
 - ▶ Show that a loan market can experience credit rationing *in equilibrium*.
- ▶ **Traditional Explanations** (which this paper challenges):
 - ▶ *Short-term:* Temporary disequilibrium due to sticky interest rates after an exogenous shock.
 - ▶ *Long-term:* Government constraints (e.g., usury laws).
- ▶ **The New Paradigm:**
 - ▶ Rationing is a natural outcome of **imperfect information**, not external constraints.

Definition

Credit rationing

1. Among homogeneous applicants some of them receive the loan while others do not, and the rejected applicants would not receive the loan even if they offered higher interest rate
2. There are identifiable groups who, with given supply of credit, are unable to take out a loan at any interest rate, even though with a larger supply of credit, they would

Model Setup



- **Default condition:** $C + R \leq B(1 + \hat{r})$
- Borrower Net Return: $\pi(R, \hat{r}) = \max(R - (1 + \hat{r})B; -C)$
- Bank Return: $\rho(R, \hat{r}) = \min(R + C; B(1 + \hat{r}))$

Model Setup: Borrowers and Projects

- **Project choice:** Borrower chooses the project θ , which has fixed cost, is not divisible and its realization requires taking out a loan.
- **Returns:** The return R is a random variable with cumulative distribution $F(R, \theta)$ and density $f(R, \theta)$

Risk

Greater θ corresponds to greater risk i.e. for $\theta_1 > \theta_2$, if

$$\int_0^\infty Rf(R, \theta_1)dR = \int_0^\infty Rf(R, \theta_2)dR$$

then

$$\forall y \geq 0 \quad \int_0^y F(R, \theta_1)dR \geq \int_0^y F(R, \theta_2)dR$$

Model Simplifications & Assumptions

Banks

- ▶ **Risk Preferences:**
Lenders are risk-neutral.
- ▶ Want to identify borrowers who are most likely to repay.
- ▶ Banks compete with each other via their choice of interest rate.
- ▶ The supply of loanable funds is *not* affected by the interest rate the bank charges.

Borrowers

- ▶ **Risk Preferences:**
Borrowers are risk-neutral.
- ▶ The borrower has a given amount of equity.
- ▶ The amount borrowed for every project is identical.
 - ▶ *Implication:* The distribution of loan applications is identical to the distribution of monetary demand.

Interest rate as a screening device (1/2)

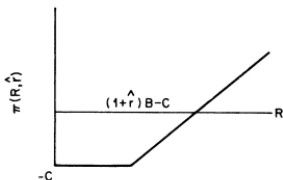
- For $\hat{r}$ there is a critical value $\hat{\theta}$ such that a firm borrows iff $\theta > \hat{\theta}$:

$$\Pi(\hat{r}, \hat{\theta}) = \int_0^{\infty} \max[R - (\hat{r} + 1)B; -C] dF(R, \hat{\theta}) = 0$$

- As the interest rate $\hat{r}$ increases, the critical threshold $\hat{\theta}$ for loan applicants also increases.

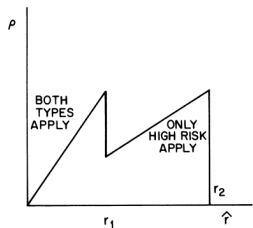
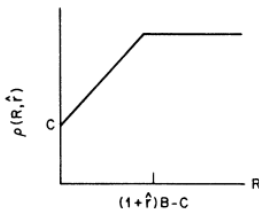
Using the implicit function and envelope theorems:

$$\frac{d\hat{\theta}}{d\hat{r}} = -\frac{\frac{\partial \Pi}{\partial \hat{r}}}{\frac{\partial \Pi}{\partial \hat{\theta}}} = \frac{B \int_{(1+\hat{r})B-C}^{\infty} dF(R, \hat{\theta})}{\frac{\partial \Pi}{\partial \hat{\theta}}} > 0$$



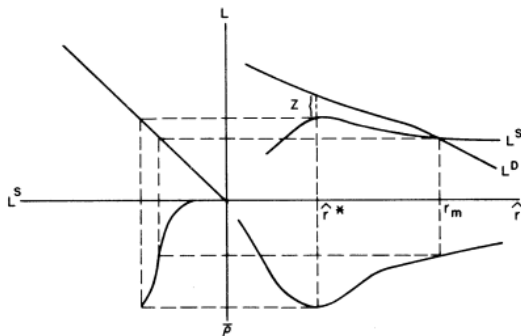
Interest rate as a screening device (2/2)

- ▶ The expected return on a loan to a bank is a **decreasing function** of the loan's riskiness.
- ▶ If there are a discrete number of potential borrowers each with a different θ , $\bar{\rho}(\hat{r})$ will **not** be a monotonic function of r .
 - ▶ As each successive group drops out of the market, there is a discrete fall in $\bar{\rho}$.



Equilibrium with Credit Rationing (1/2)

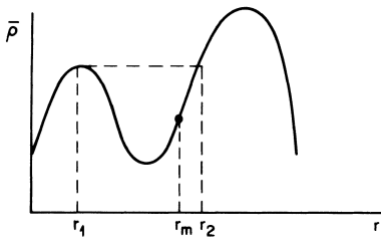
- Whenever $\bar{\rho}(r)$ has an **interior mode** (a distinct maximum point), there exist supply functions of funds such that competitive equilibrium entails credit rationing.



Equilibrium with Credit Rationing (2/2)

If the $\bar{\rho}(r)$ function has **several modes**, market equilibrium could either be characterized by:

- ▶ a **single interest rate** at or below the market-clearing level, or
- ▶ by **two interest rates**, with an excess demand for credit at the lower one.



Interest Rate as an Incentive Mechanism (1/2)

- ▶ **Intuition:** Increasing the interest rate **increases the relative attractiveness of riskier projects**, for which the return to the bank may be lower
- ▶ If at a given nominal interest rate r , a risk-neutral firm is indifferent between two projects, an increase in the interest rate results in the firm preferring the project with the higher probability of bankruptcy.

$$\pi_i(R_i, \hat{r}) = \mathbb{E}\{\max(R_i - (1 + \hat{r})B; -C)\}$$

$$\frac{d\pi_i}{d\hat{r}} = -B(1 - F_i[(1 + \hat{r})B - C])$$

Interest Rate as an Incentive Mechanism (2/2)

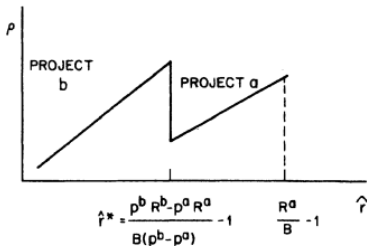
- ▶ The expected return to the bank is lowered by an increase in the interest rate at $\hat{r}$ if, at $\hat{r}$, the firm is indifferent between two projects j and k with distributions $F_j(R)$ and $F_k(R)$, j having a higher probability of bankruptcy than k , and there exists a distribution $F_l(R)$ such that
 - (a) $F_j(R)$ represents a mean preserving spread of the distribution $F_l(R)$, and
 - (b) $F_k(R)$ satisfies a first-order dominance relation with $F_l(R)$; i.e., $F_l(R) > F_k(R)$ for all R .

Example (1/2)

- ▶ All firms are identical and choose between two projects. If successful, they yield:
 - ▶ **Project A (Risky):** Return R_a , probability p_a
 - ▶ **Project B (Safe):** Return R_b , probability p_b
- Assume: $R_a > R_b$ and $p_a < p_b$.
- ▶ At interest rate $\hat{r}^*$, the firm is indifferent between A and B:

$$[R_a - (1 + \hat{r})B]p_a = [R_b - (1 + \hat{r})B]p_b$$

$$B(1 + \hat{r}) = \frac{p_b R_b - p_a R_a}{p_b - p_a} = B(1 + \hat{r}^*)$$



Example (2/2)

- ▶ The maximum expected return to a bank occurs when the bank charges an interest rate $\hat{r}^*$ if and only if
$$p_a R_a < \frac{p_b(p_b R_b - p_a R_a)}{p_b - p_a}$$
- ▶ Whenever $p_b R_b > p_a R_a$, $1 + \hat{r}^* > 0$, and ρ is not monotonic in r , so there may be credit rationing.
- ▶ As a result, banks prefer to deny loans rather than raise rates, leading to credit rationing.

Summary

Key Takeaways

- ▶ In equilibrium, a loan market may be characterized by **credit rationing**.
- ▶ The interest rate a bank charges may itself affect the riskiness of the loans by either:
 1. The **adverse selection** effect.
 2. The **incentive** effect.

Contribution

The first theoretical justification for *true* credit rationing.

Applications

Applicable to other Principal-Agent problems (e.g., agriculture).

Limitations

Primarily relies on a single-period model.